

The Dawn of Generative Artificial Intelligence in Chemistry Education

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ABSTRACT: The emergence of generative artificial intelligence has precipitated a wide range of predictions for its impact on science and the teaching and learning of science. Chemists and chemistry educators have been exploring the possibilities of this new technology, essentially as soon as it was broadly released for use. Because this technology is rapidly developing, the insights that can be gained from the work of early adopters and investigators can play a particularly important role in shaping the future of how chemistry education adapts to, and perhaps embraces, this new technology. This Special Issue provides a snapshot at this relatively early moment in the evolution of this technology for use in teaching and learning chemistry. The interplay between technology development and implementation in the classroom will always be important, but seldom more so than in the early stages of work. The set of articles presented in this Special Issue (<https://pubs.acs.org/page/jceda8/vi/genai2024>) shows a robust range of options for use, and potentially for misuse, and can help guide those interested in finding ways to enhance the emerging learning opportunities afforded by this new tool.

The COVID-19 pandemic profoundly shook the higher education sector, including chemistry education,^{1,2} forcing a rapid shift to online learning and exposing vulnerabilities in traditional educational models. Just as institutions began adapting to this new normal, the emergence of generative artificial intelligence (AI) presented a new set of challenges and opportunities.

As dawn breaks on generative AI, it heralds a new era in chemistry education, pushing the traditional boundaries and opening up new pathways for pedagogical innovation. This ground-breaking technology is capable of creating content, solving problems, and engaging in sophisticated dialogue. Akin to a digital alchemist, it has the potential to “transmute” traditional teaching methodologies into more interactive experiences and revolutionize how chemistry is taught and learned. Its arrival has already catalyzed significant innovation, illustrated by the corpus of articles in this Virtual Special Issue (<https://pubs.acs.org/page/jceda8/vi/genai2024>). However, this new dawn is not without its shadows. The rapid integration of AI raises ethical concerns about academic integrity, apprehensions about inaccuracies and bias in AI-generated content, and cybersecurity risks.

From the start of 2023, the *Journal* has seen a steady increase of submitted manuscripts describing the effect of generative AI arrival on chemical education. The call for papers in September 2023 solicited manuscripts about the implementation and impact of generative AI on educational practice and research.³ In that solicitation, we encouraged descriptions of investigations that consider appropriate uses of the technology and how such uses can, and do, contribute to the teaching and learning of chemistry. This call for papers helped shape the types of work authors submitted to the *Journal*, such that broadly speaking, we have seen three overlapping waves of manuscripts.

1. Initial reports often focused on generative AI capabilities in terms of what it can (or cannot) do.

2. A second wave included more manuscripts where educators' views on and actions with generative AI (How can/do instructors use it?) were explored.
3. A third wave placed the focus on students' perceptions of and practices/uses of generative AI (How can/do students use it?).

These three phases as well as conceptual papers⁴ have established a growing body of literature that allows the formulation of a conceptual framework for interpreting the multifaceted influence of generative AI on chemical education. The CATALyST framework provides a structure for such interpretation through seven key dimensions (Figure 1).

Context

Generative AI is utilized in diverse educational settings, including laboratories,^{5,6} classrooms,⁷ and assessment environments.^{8–11} These contexts can leverage AI's unique capabilities to provide transformative opportunities and enhance educational experiences.

Applications

The practical uses of generative AI in education are ranging from the preparation of teaching materials (preclass/lesson planning)^{6,12} to facilitating in-class problem solving⁷ to assisting with lab report writing after the class.^{13,14}

Technology

The rapid advancement of AI technology significantly influences chemical education. After early reports about AI limitations for

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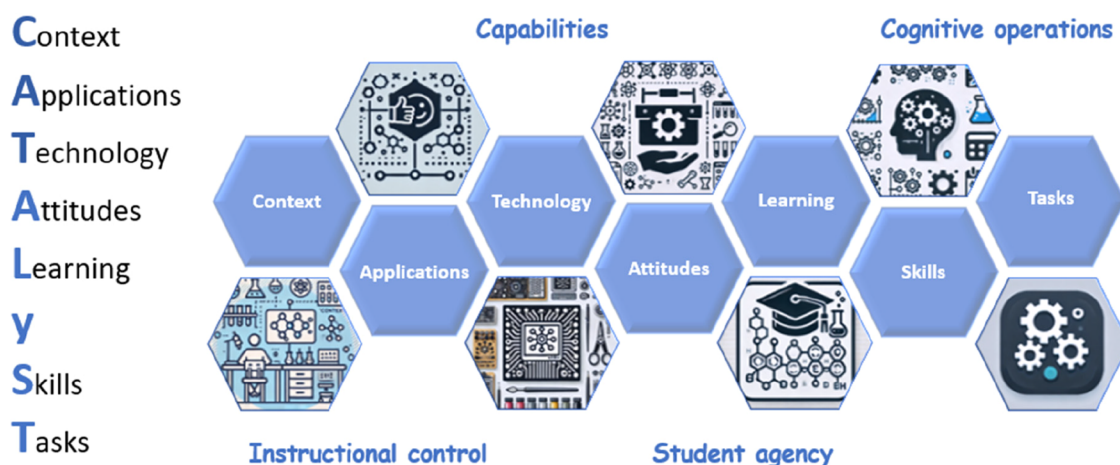


Figure 1. CATALyST conceptual framework. In this framework, there are seven dimensions (blue hexagons) grouped into four domains (blue text labels). The illustrative hexagons were created with the assistance of DALL·E.

doing and reasoning about chemistry,^{5,8–10,13,15–17} instructors and researchers are now reckoning with and adapting to emerging AI capabilities, such as reasoning about organic reaction mechanisms^{18,19} or multimodality, i.e., the ability to handle visual inputs.²⁰

Attitude

Understanding the perceptions, acceptance levels, and ethical concerns of students and educators is crucial for the successful integration of AI.^{11,13,14,21–24} These attitudes and AI literacy shape how AI is adopted and applied in educational contexts.

Learning

The impact of generative AI on students' learning experiences, outcomes, and knowledge acquisition is profound. AI-driven tools, such as curated Chatbots,²⁵ provide personalized and adaptive learning experiences.

Skills

The use of generative AI necessitates the deployment of generic skills, such as critical thinking,^{7,11,14,21,24} and development of new skills, such as prompt engineering^{24,26} and AI literacy.²³ These skills are essential for effectively interacting with AI systems and harnessing their full potential.

Tasks

Educational activities involving AI, such as coding²⁷ and essay or lab report writing,^{11,13,14,21,22,24} require intellectual engagement and critical thinking.

Within the CATALyST framework, certain concepts can be grouped into domains, based on their underlying relationships. Context and Applications fall under Instructional Control, reflecting the structured environments and practical uses of AI. Attitude and Learning are governed by Student Agency, emphasizing the importance of perceptions and active engagement. Skills and Tasks are categorized as Cognitive Operations, highlighting the intellectual engagement and skill development required for and fostered by AI tools.

Insights from the literature so far reveal that the applications of generative AI in teaching and learning are diverse and impactful, with stakeholders showing a range of perceptions toward its integration, from cautiously positive to critical and apprehensive. The impact of AI on cognitive operations, particularly on deploying and developing skills and completing tasks, remains an area ripe for further exploration. Emerging

research questions include those addressing how students use generative AI and whether prompt engineering is a specialized skill or simply a refined form of questioning and interaction. Innovative teaching strategies, such as AI-assisted personalized learning and collaborative AI-human activities/research projects, promise to further transform chemistry education, paving the way for a future where artificial and human intelligences coexist and thrive in the educational space.

As is true of all special issues published in the *Journal*, significant time and effort are associated with collecting a broad and diverse set of scholarly work. Accordingly, we thank everyone who submitted manuscripts for this Virtual Special Issue as well as all the reviewers who provided their time and shared their expertise to improve the manuscripts and to make this collection a valuable resource for the readers. The handling editors themselves learned a lot about various opportunities to apply generative AI in chemistry curriculum as well as various views about both benefits and risks of this relatively new and constantly developing technology. We hope readers of this Virtual Special Issue will find the collection enlightening and practically useful. We understand that this collection of manuscripts represents just the beginning of the journey that the chemical education community has embarked upon in the end of 2022 when ChatGPT burst onto the scene. We encourage all members of the community to engage or continue engaging with generative AI for the betterment of their students' learning. We hope that this collection will help them in this endeavor, and we are looking forward to members sharing their experience and findings.

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Notes

Views expressed in this editorial are those of the authors and not necessarily the views of the ACS.

Elizabeth Yuriev is an Associate Professor in the Faculty of Pharmacy and Pharmaceutical Sciences, Monash University. She is an experienced chemistry educator and education

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Thomas A. Holme is a Morrill Professor in the Department of Chemistry, Iowa State University. His research has two distinct strands: chemical education research and human–computer interaction. He served as Director of the Examinations Institute of the American Chemical Society from 2002 to 2015 and conducts education research to improve the quality of information that can be obtained from exams and other forms of assessment. In addition, he has developed and implemented curricular materials to introduce systems thinking concepts and applications of artificial intelligence in General Chemistry.

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